A Network Analysis of NSF-Funded Interdisciplinary Research

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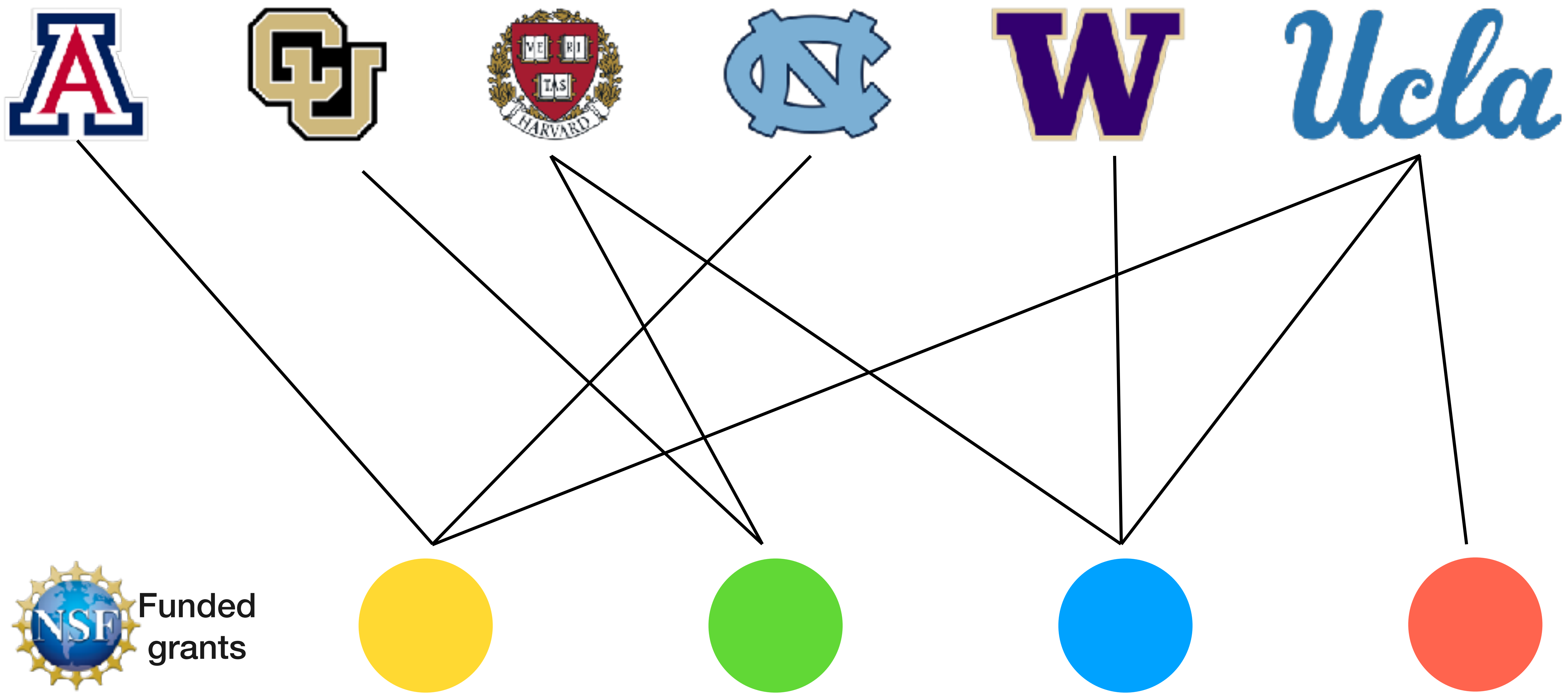
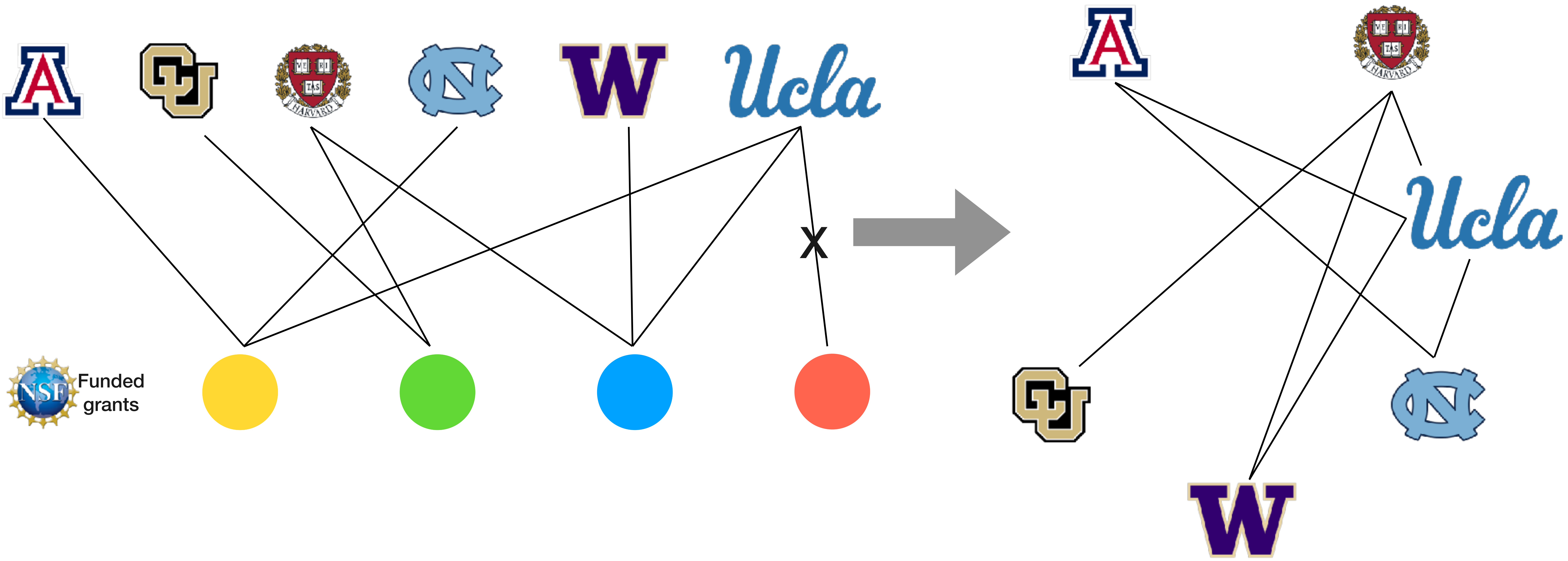
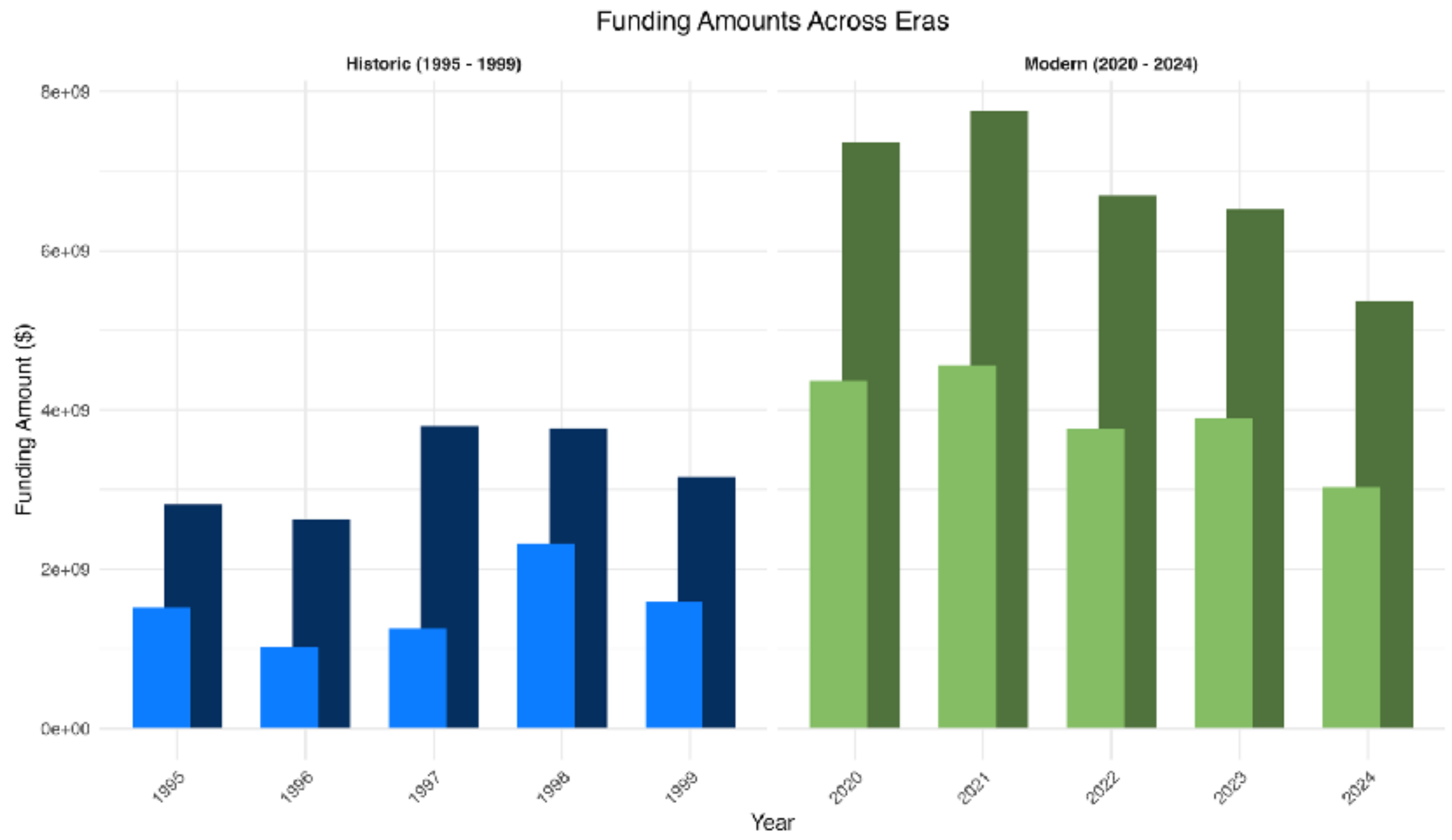
Bipartite Network Construction

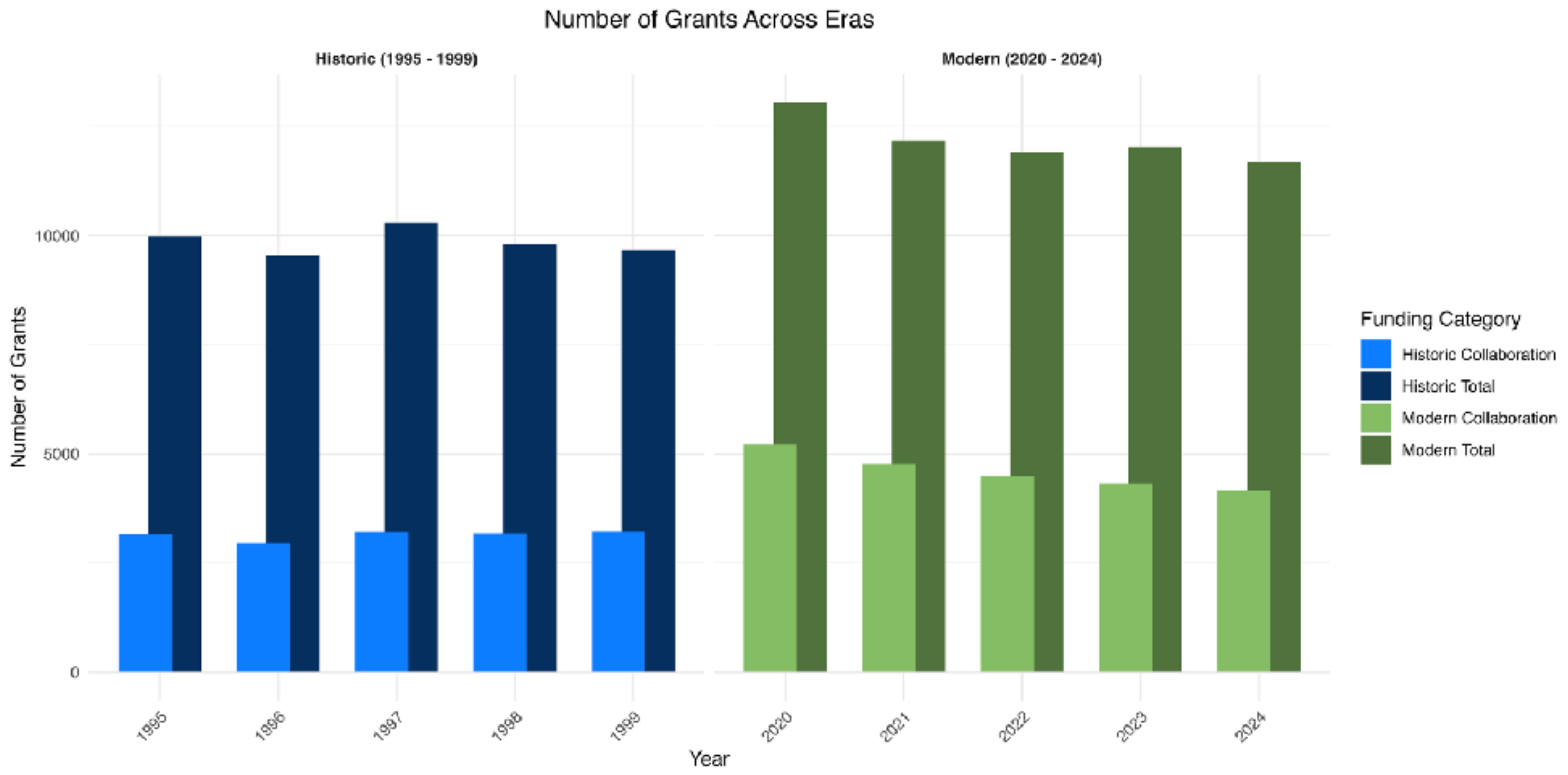


Bipartite Network Construction

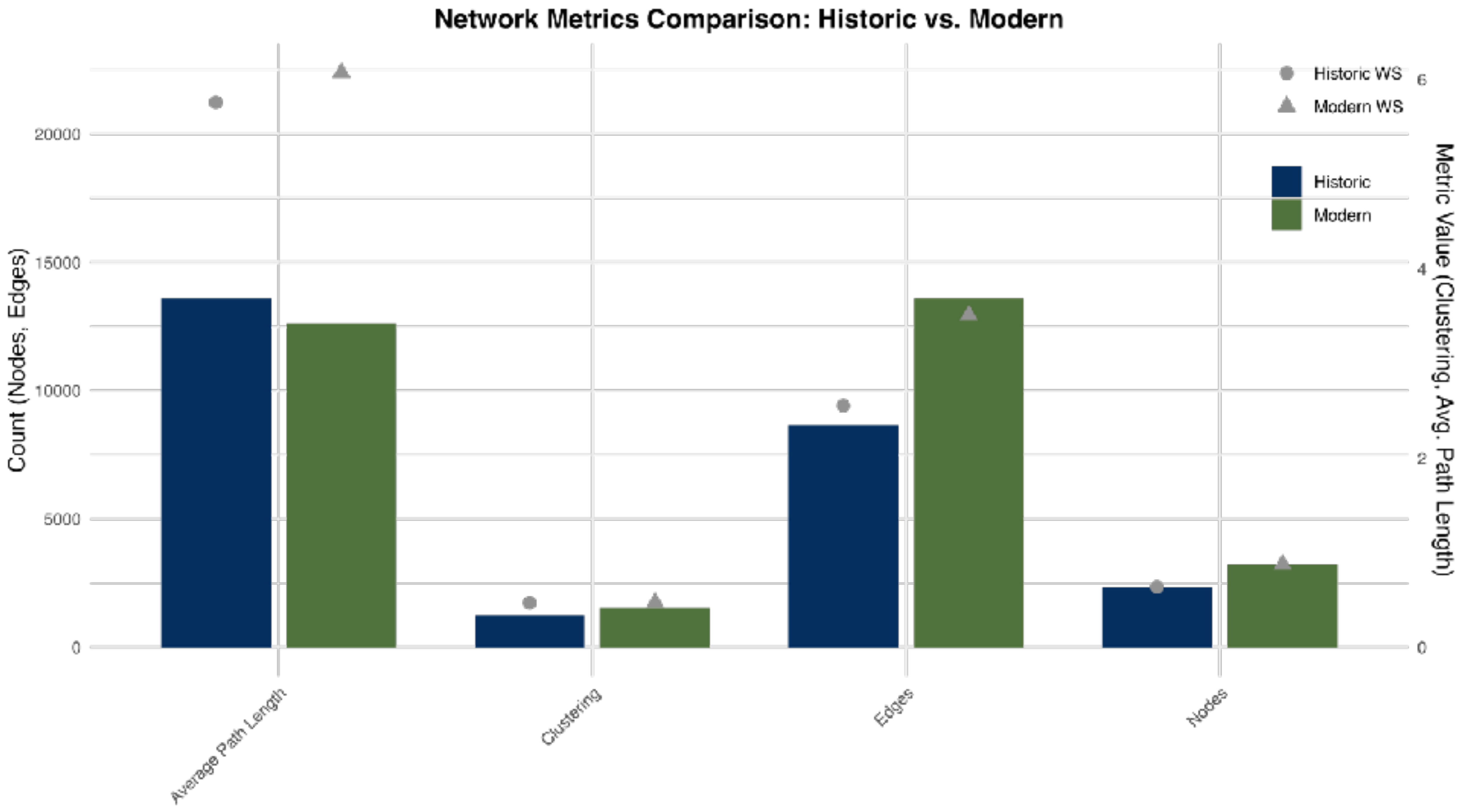


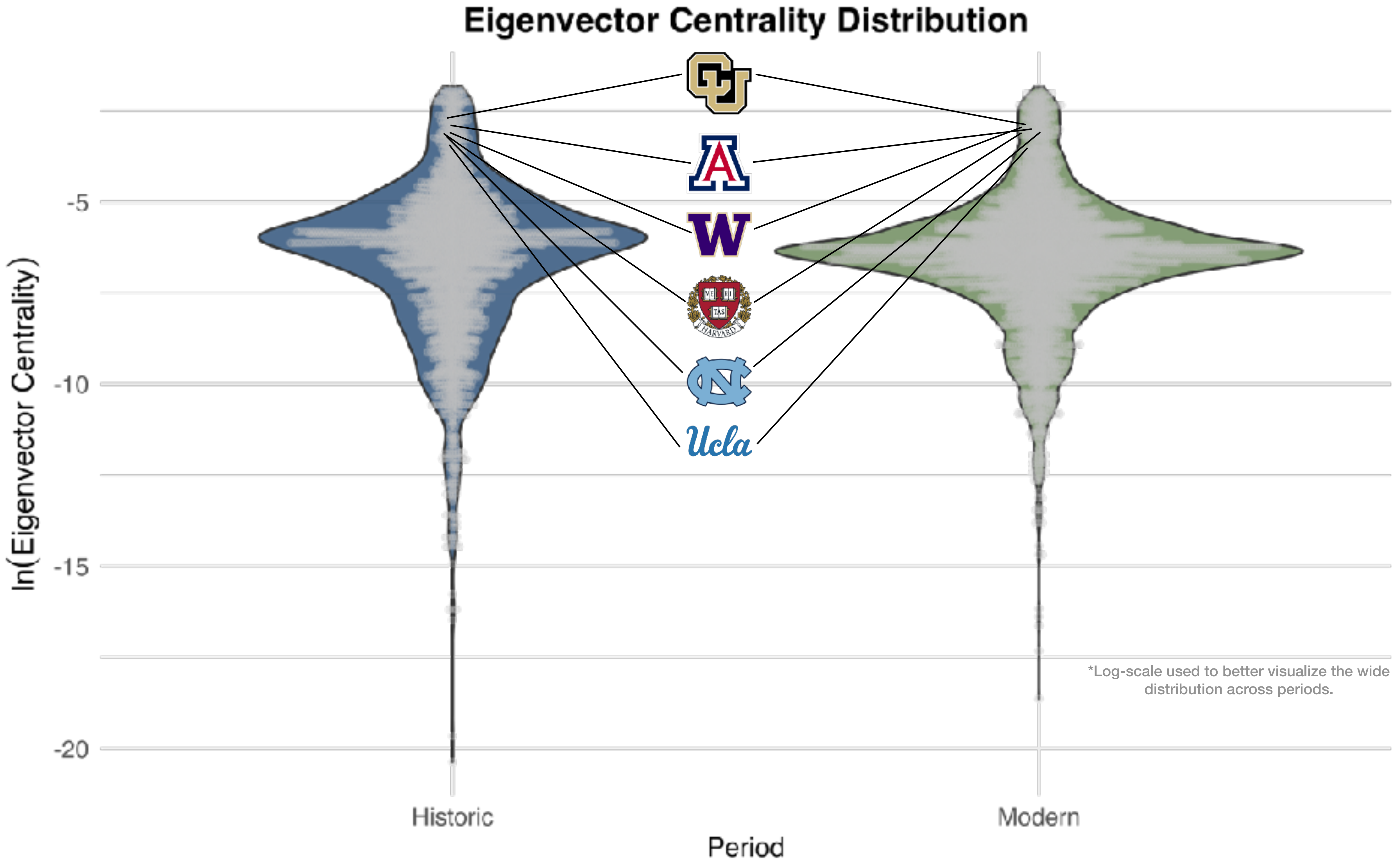
Number and Amount of Grants





Results

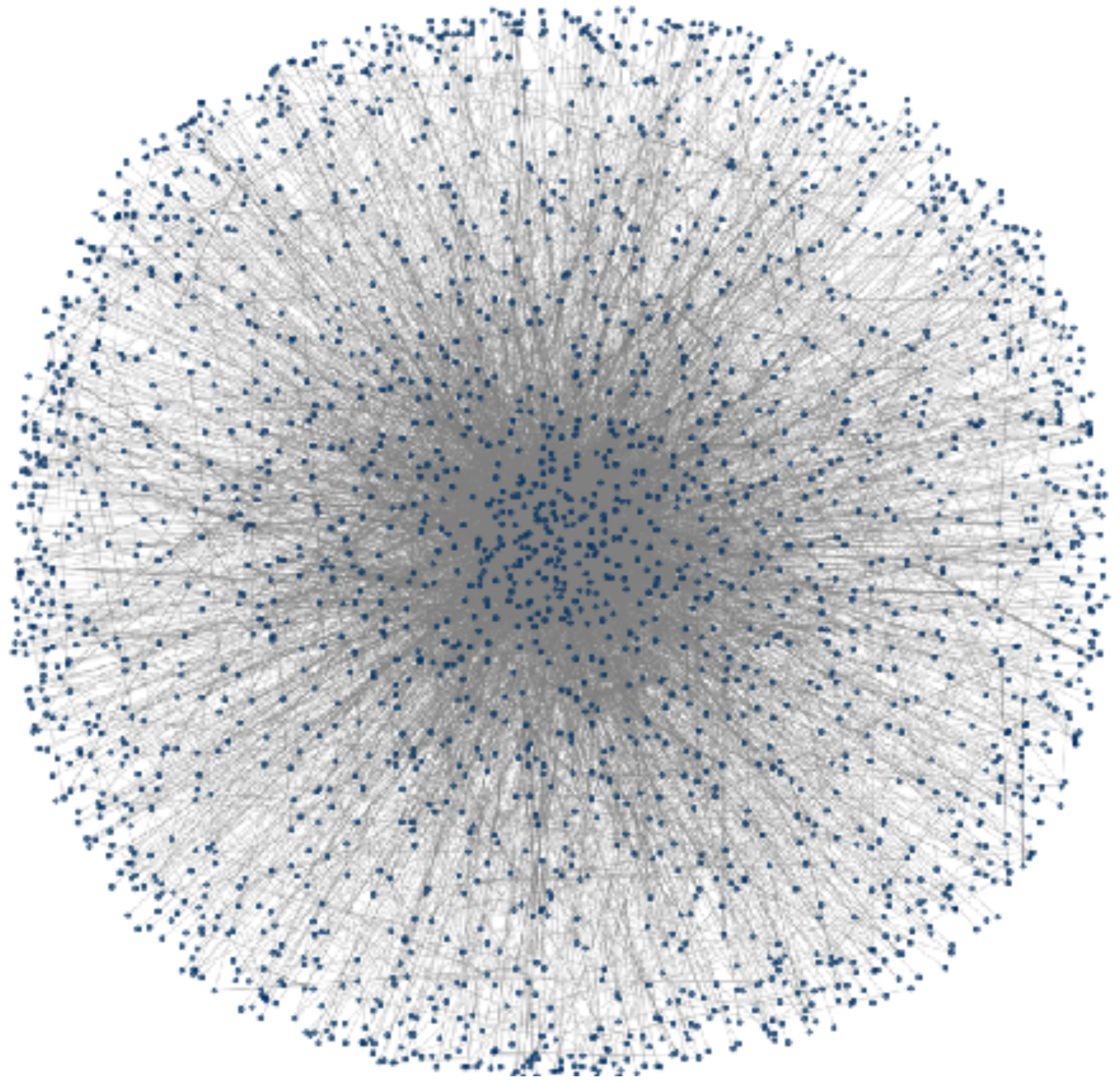

Network Metrics Comparison: Historic vs. Modern
Count (Nodes, Edges)
Metric Value (Clustering, Avg. Path Length)
Historic WS
Modern WS
Historic
Modern
Average Path Length
Clustering
Edges
Nodes


Eigenvector Centrality Distribution
ln(Eigenvector Centrality)
Historic
Modern
Period
*Log-scale used to better visualize the wide distribution across periods.

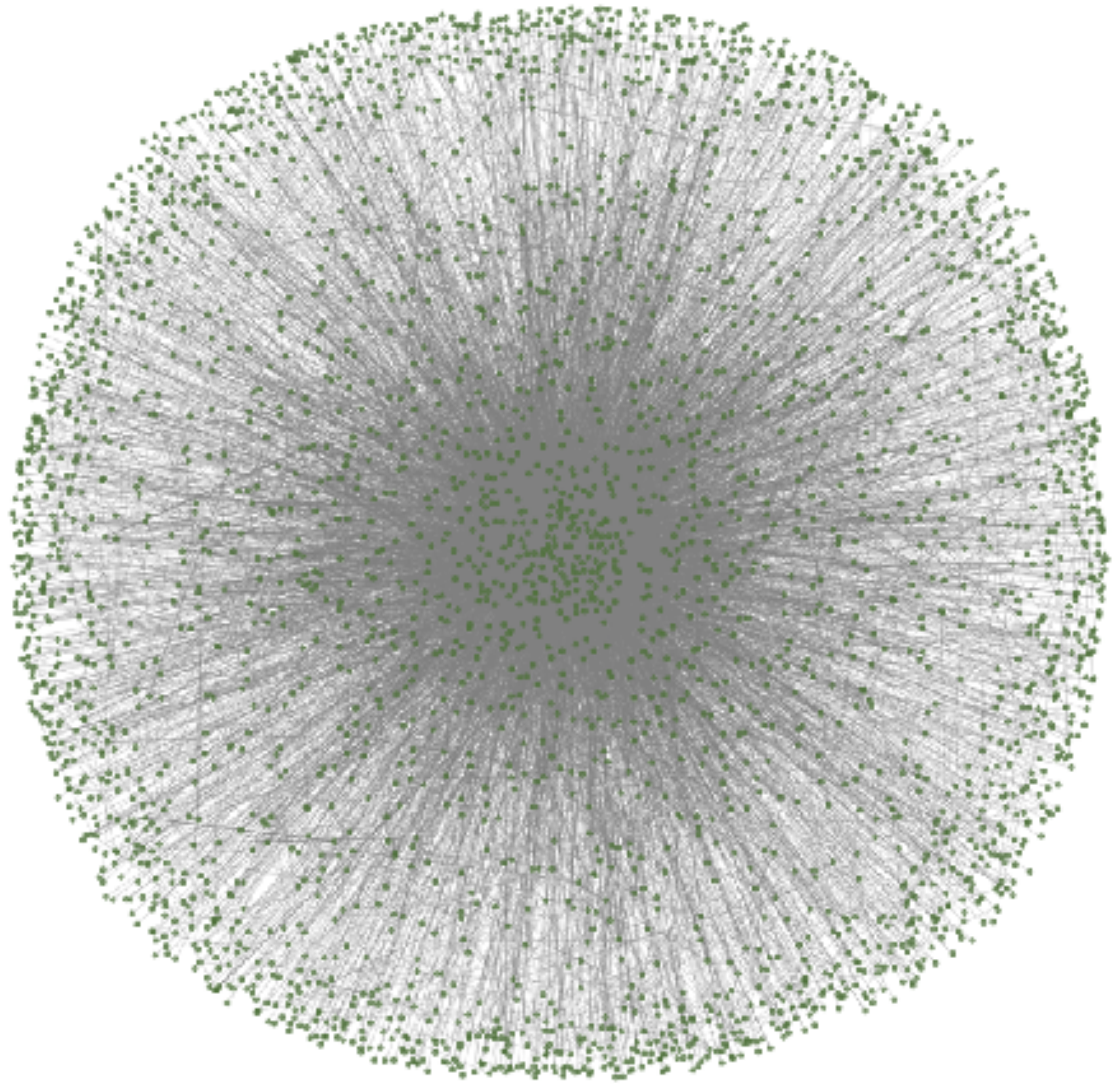
Conclusions

- **Modern network shows greater connectivity** supported with WS null models
- **Higher clustering** in the modern network suggests denser collaboration
- **Shorter path lengths** in the modern network indicate increased interconnectivity between institutions
- **Eigenvector centrality is more widely distributed** in the modern network ($p<0.001$, levene test), indicating the influence is shared across a broader range of institutions
- **More institutions hold high centrality**, suggesting shifts in influence and potential broadening of key collaborators

Historic Network
(1995-1999)



Modern Network
(2020-2024)



Thanks, Questions?